

Pranav Vishwanatham

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EDUCATION

Master of Engineering in Computer Science and Applications, GPA - 3.87/4

January 2023 – December 2024

• **Virginia Polytechnic Institute and State University (Virginia Tech)**

Blacksburg, Virginia

Master of Technology in Software Engineering (Integrated), GPA - 8.09/10

July 2016 – May 2021

• **Vellore Institute of Technology (VIT University)**

Vellore, India

SKILLS

- **Programming & Web Development:** Python, C/C++, Embedded C, HTML/CSS, JavaScript, ReactJS, VueJS, D3.js, MySQL.
- **Data Science & Advanced ML Technologies:** Machine Learning (Numpy, Pandas, Scikit-learn), Deep Learning (Pytorch, Tensorflow)
- **Tools:** DOORS, ALM, Jira, ETAS (ASCET), Matlab/Simulink, TestIDE, SharCC, Jenkins, Splunk, Tableau, Figma, SVN, GIT, MongoDB.
- **Software Practices:** SDLC(V-Model), Agile (Scrum), Requirements Engineering, Software Design & Documentation, Software Development, Software Review, Software Metrics, Software Testing (Unit Testing, Component Testing/SIL, HIL), Software Integration.
- **Automotive:** Embedded Software Programming, Model-Based Design(MBD), Vehicle Dynamics, Braking Systems, AUTOSAR, Automotive SPICE, CAN Protocol, FMEA, ISO 26262.

EXPERIENCE

Associate Software Engineer

September 2021 – December 2022

Bosch Global Software Technologies

Hyderabad, India

- Managed Application Software (ASW) development for braking system ECUs within Chassis Controls division, focusing on safety critical automotive control systems projects such as Electronic Stability Program (ESP) and Integrated Power Brake (IPB) for the Volkswagen (VW) Group.
- Coordinated embedded systems programming for Cooperative Regenerative Braking (CRB), Driver Brake Request (DBR) and Sensor Signal Processing (SSP) functions across ESP and IPB projects.
- Analyzed customer and system requirements to derive functional requirements, executed software design changes, and implemented solutions using Model-Based development with MATLAB/Simulink and ETAS ASCET, ensuring alignment with the V-Model lifecycle for verification and validation, in compliance with AUTOSAR standard, ASPICE processes, and MISRA coding guidelines.
- Conducted code reviews and implemented unit testing, Software-in-Loop testing (SIL), Hardware-in-Loop (HIL) testing by creating various test cases to meet ISO26262 functional safety standards before software delivery, while also working on integrating the software seamlessly.
- Collaborated within cross-functional Agile teams using Scrum, actively participating in sprint planning, backlog grooming, and daily stand-ups ensuring timely delivery of software components.
- Secured the Function Owner position within six months by demonstrating exceptional proficiency, leading the CRB and DBR teams to achieve a 20% increase in positive customer feedback and a 10% reduction in defect leakages.
- Recognized for delivering results by coordinating team efforts, earning the Best Newcomer award for exceptional performance.

Software Engineering Internship

November 2020 – June 2021

Robert Bosch Engineering and Business Solutions Private Limited

Bangalore, India

- Created the Automated Resource Optimizer tool by gathering and analyzing data of automotive controls ECU software under ESP braking systems project to identify patterns and optimal resource usage configuration.
- Tuned parameters to execute Complete Software (CSW) builds for ESP systems using the Jenkins CI/CD pipeline, flashed the embedded software, and ran Lab Car testing scripts to collect runtime metrics for resource usage and performance evaluation.
- Integrated Python script into pipeline to create CSV file and generate HTML reports with graph visualizations of resource utilization data.
- Applied machine learning algorithms to predict optimal resource configurations based on gathered data, achieving 10% reduction in resource usage and deployed the solution using Python (Tkinter) and MongoDB to enhance interface efficiency and streamline optimization processes.
- Earned full-time position within the team as an Associate Software Engineer based on my exceptional performance and demonstrated skills.

PROJECTS

EasyHire

- Developed a MERN stack hiring solution to streamline recruitment workflows, reducing time-to-hire by 25% through efficient job posting and application management.

Picture captioning

- Designed an image captioning model using CNN and LSTM layers in TensorFlow with ImageNet pre-trained weights, achieving BLEU scores up to 60.00 across various test cases.

AlertAI

- Implemented an AI-powered tool using Twilio, Azure Speech-to-Text, GPT-4, and MongoDB to assist 911 dispatchers with real-time call transcription, data extraction, and decision-making via an interactive dashboard.